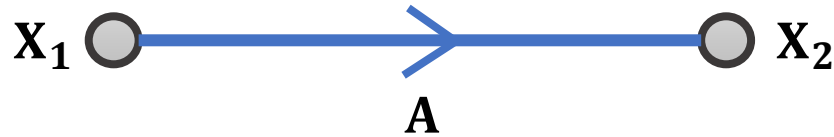


Signal Flow Graph Terms and Rules

Terms of Signal Flow Graph

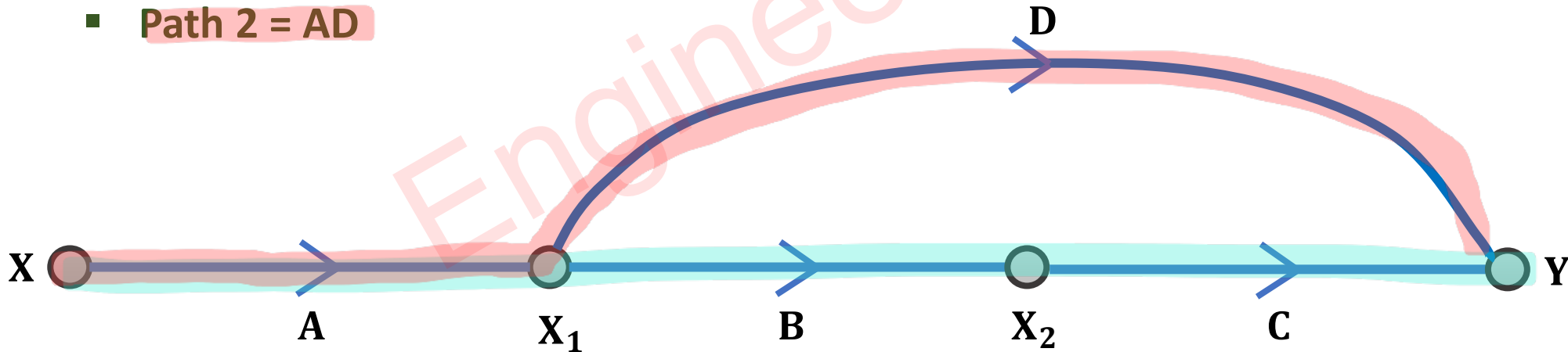
□ Input Node X_1 (Source Node) and Output Node X_2 (Sink Node)

- Output $X_2 = \text{Gain (A)} \times \text{Input } X_1$



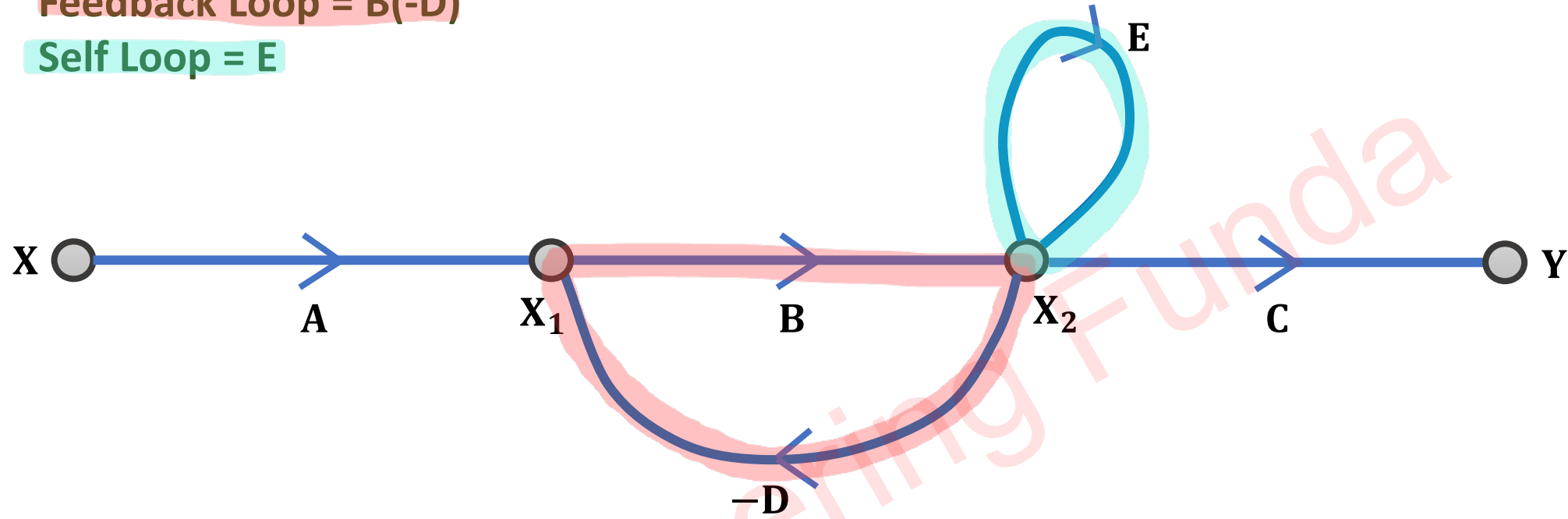
□ Forward Paths

- Path 1 = ABC
- Path 2 = AD



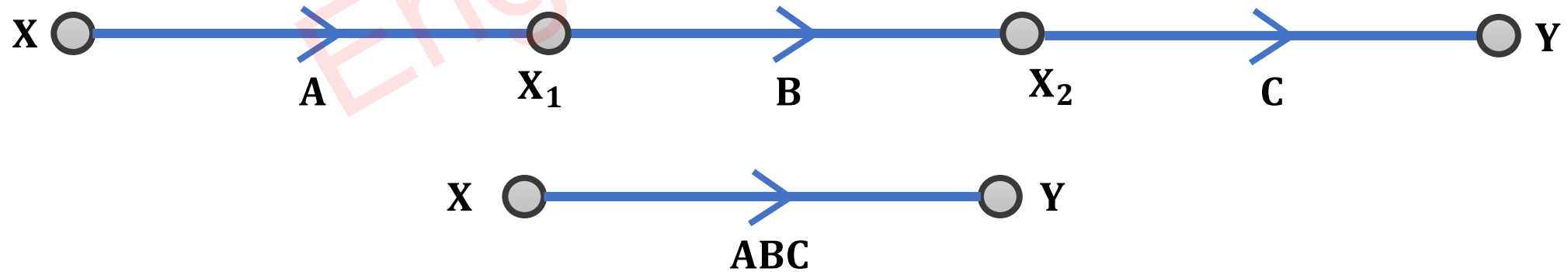
□ Feedback Loop and Self Loop

- Feedback Loop = $B(-D)$
- Self Loop = E



□ Chain Node

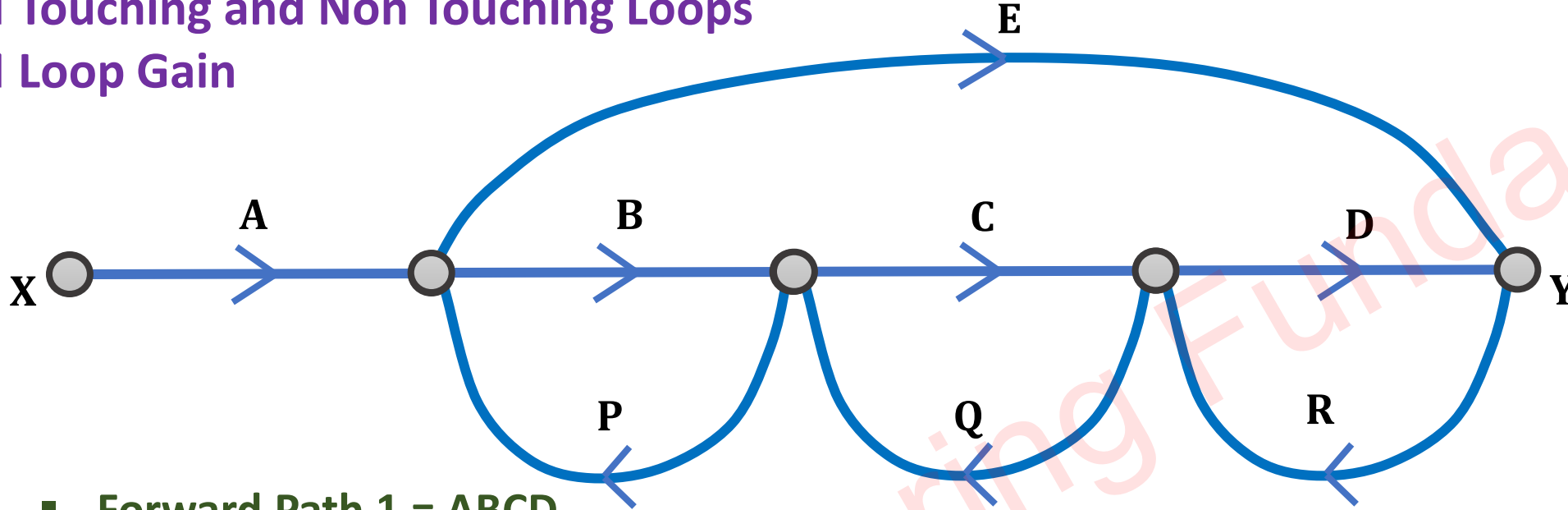
- Chain Nodes are X_1 and X_2



□ Touching Path and Non Touching Path

□ Touching and Non Touching Loops

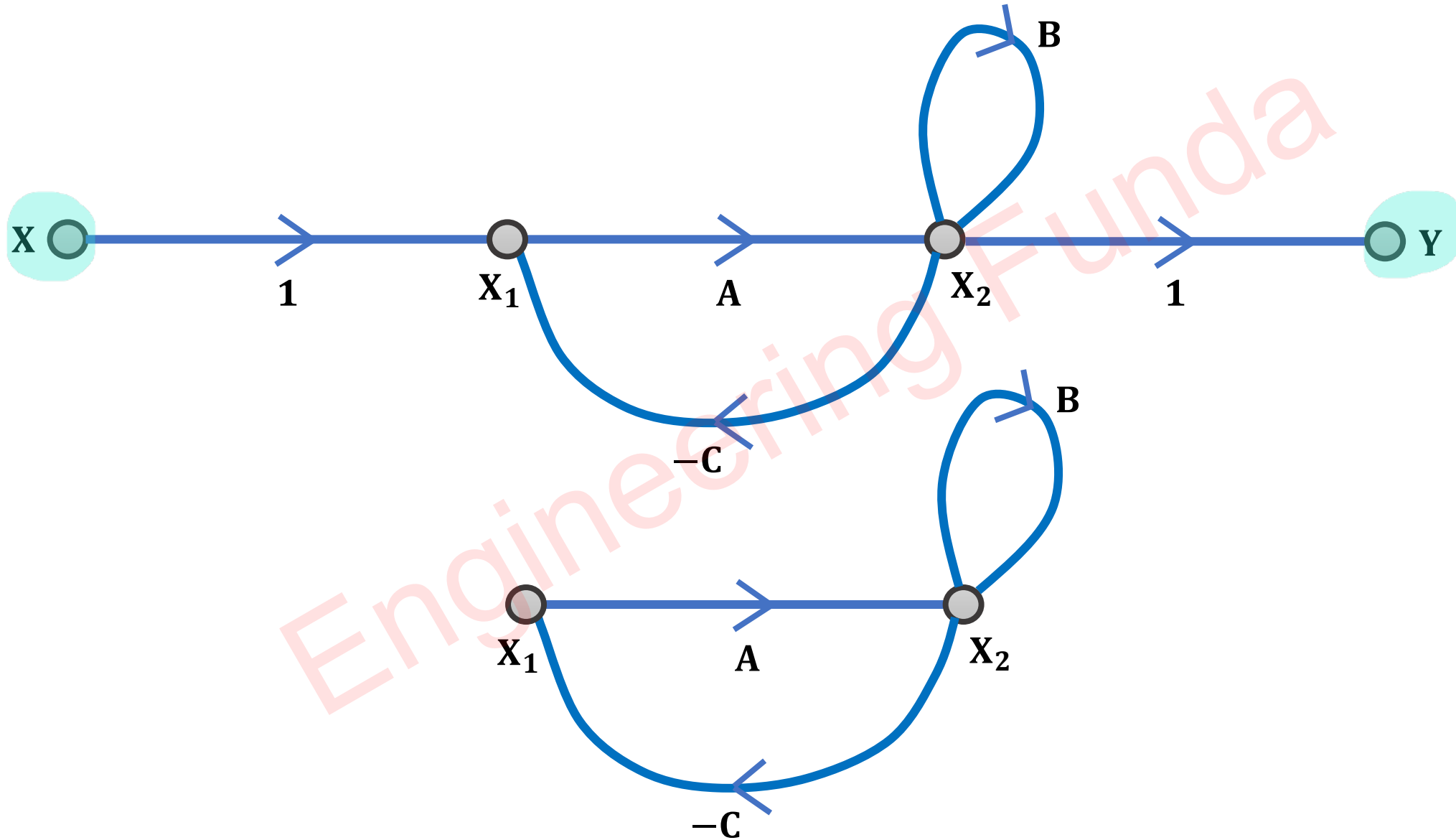
□ Loop Gain



- Forward Path 1 = ABCD
- Forward Path 2 = AE
- Loop 1 = BP {Here BP is Loop Gain}
- Loop 2 = CQ
- Loop 3 = DR
- Loop 4 = ERQP
- Forward Path 1 is Touching to All Loops.
- Forward Path 2 is Not Touching to Loop 2.
- Loop 1 and Loop 3 are Non Touching Loops.

□ Dummy Node

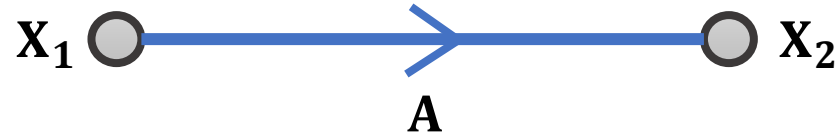
- Here Node X and Node Y are Dummy Nodes.



Rules of Signal Flow Graph

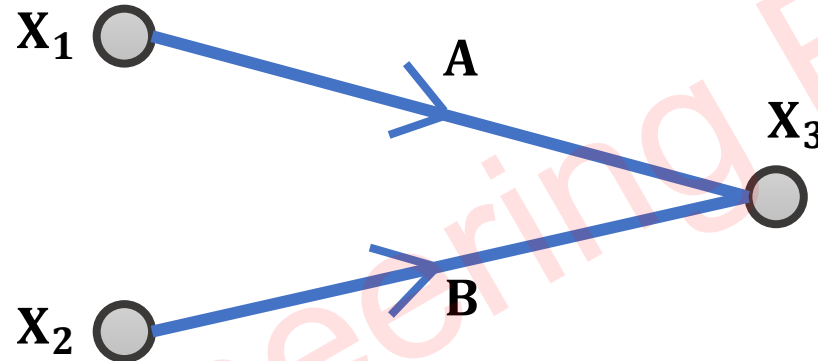
□ Rule - 1

- $X_2 = AX_1$



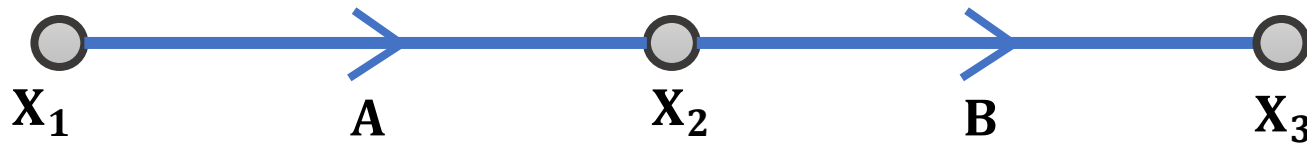
□ Rule - 2

- $X_3 = AX_1 + BX_2$



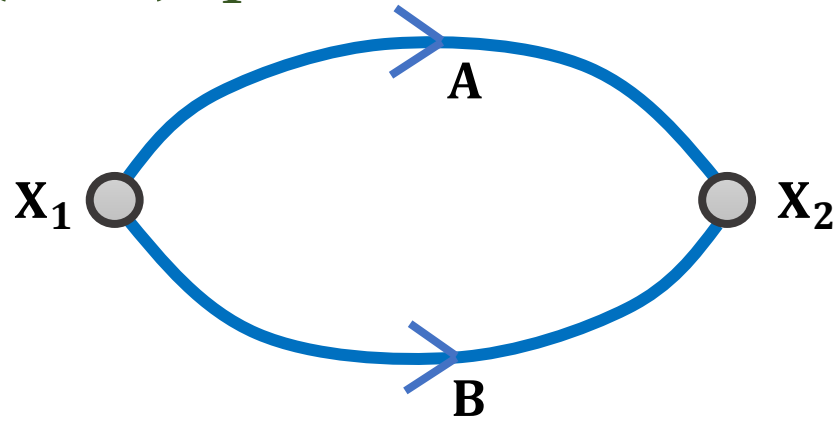
□ Rule - 3

- $X_3 = ABX_1$



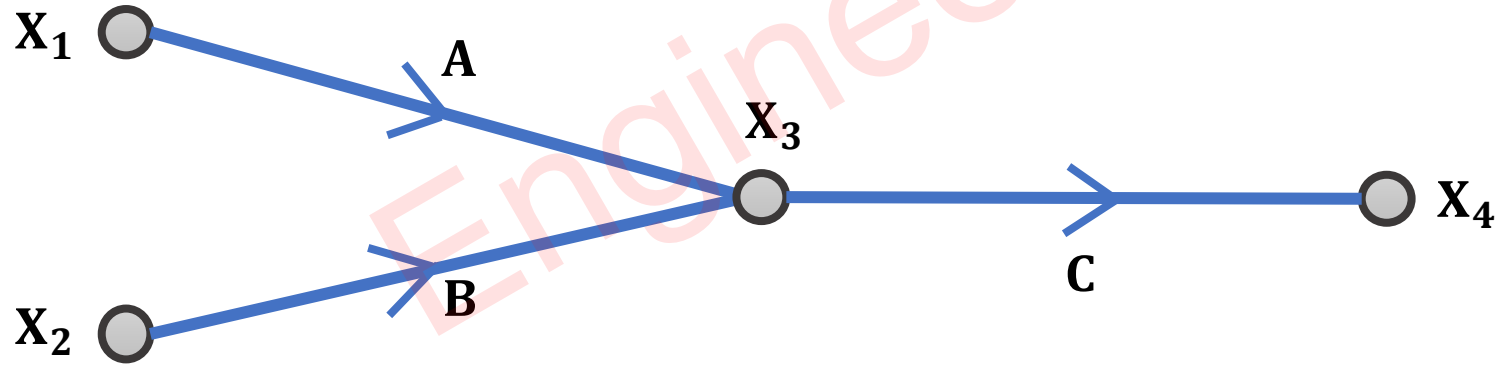
□ Rule - 4

- $X_2 = (A + B)X_1$

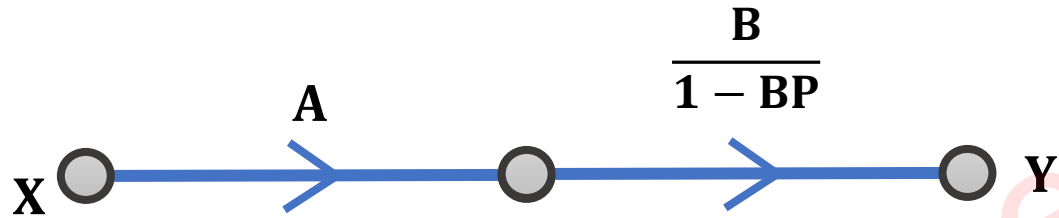
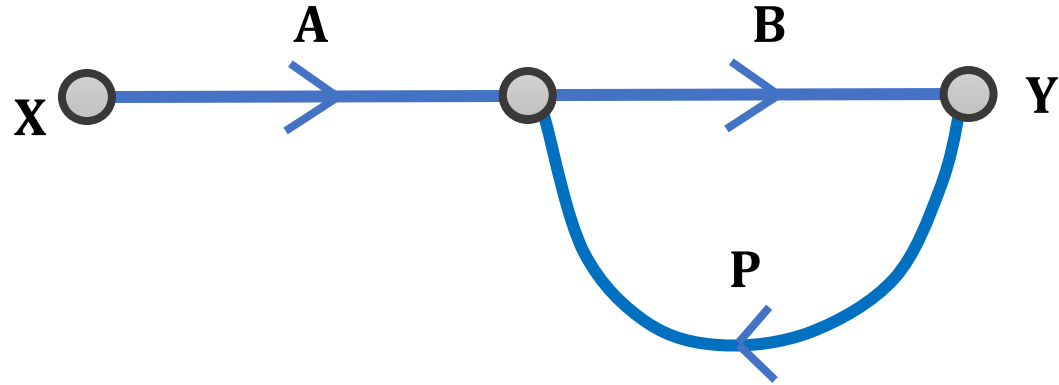


□ Rule - 5

- $X_4 = (AX_1 + BX_2)C$



□ Rule - 6



- $Y = X \left(\frac{AB}{1 - BP} \right)$

Meson's Gain Formula For Signal Flow Graph

Steps of Meson's Gain Formula

- ❑ Find Forward paths. (E.g. $F_1, F_2, \dots, F_n$)
- ❑ Find Single Loops. (E.g. $L_1, L_2, \dots, L_m$)
- ❑ Find Two Non-Touching Loops (E.g. $L'_1, L'_2, \dots, L'_p$)
- ❑ Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops} - \dots \dots \dots$$

- ❑ Find $\Delta_1, \Delta_2, \dots, \Delta_n$ based on Forward Paths

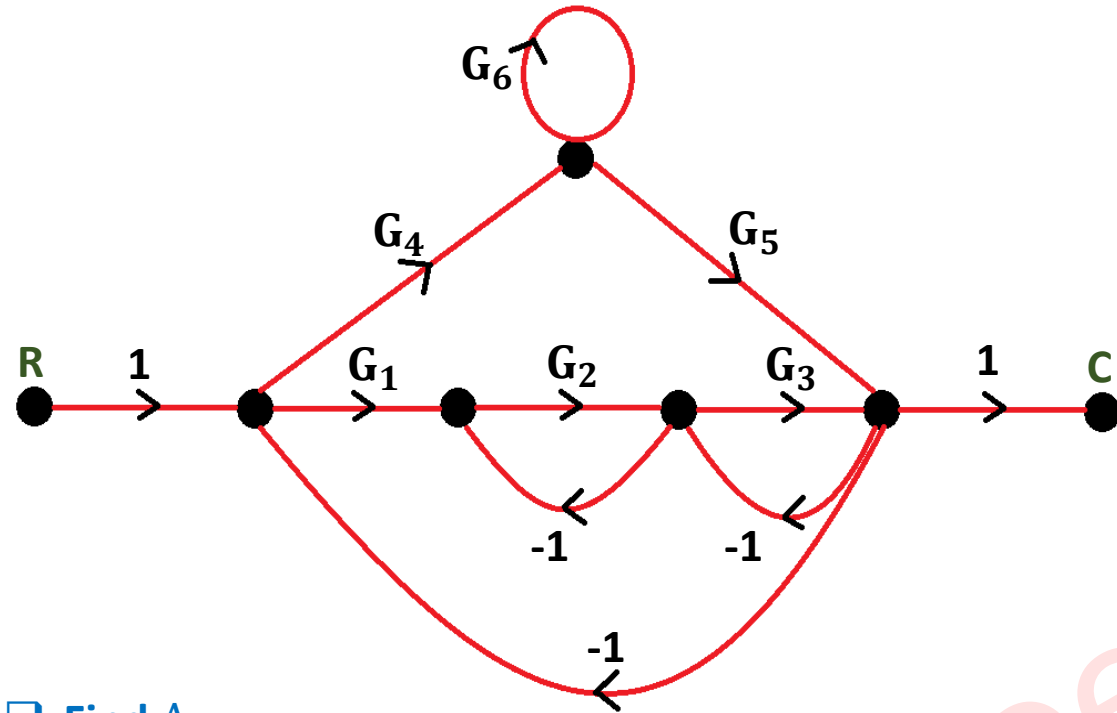
$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2$$

- ❑ Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1\Delta_1 + F_2\Delta_2 + \dots + F_n\Delta_n}{\Delta}$$

1. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = 1 \times G_1 \times G_2 \times G_3 \times 1 = G_1 G_2 G_3$$

$$\Rightarrow F_2 = 1 \times G_4 \times G_5 \times 1 = G_4 G_5$$

Find Single Loops

$$\Rightarrow L_1 = G_1 \times G_2 \times G_3 \times -1 = -G_1 G_2 G_3$$

$$\Rightarrow L_2 = G_2 \times -1 = -G_2$$

$$\Rightarrow L_3 = G_3 \times -1 = -G_3$$

$$\Rightarrow L_4 = G_6$$

$$\Rightarrow L_5 = G_4 \times G_5 \times -1 = -G_4 G_5$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_4 = -G_1 G_2 G_3 \times G_6 = -G_1 G_2 G_3 G_6$$

$$\Rightarrow L_2' = L_2 \times L_4 = -G_2 \times G_6 = -G_2 G_6$$

$$\Rightarrow L_3' = L_2 \times L_5 = -G_2 \times -G_4 G_5 = G_2 G_4 G_5$$

$$\Rightarrow L_4' = L_3 \times L_4 = -G_3 \times G_6 = -G_3 G_6$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1 - G_6$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1 + G_2$$

Find Δ

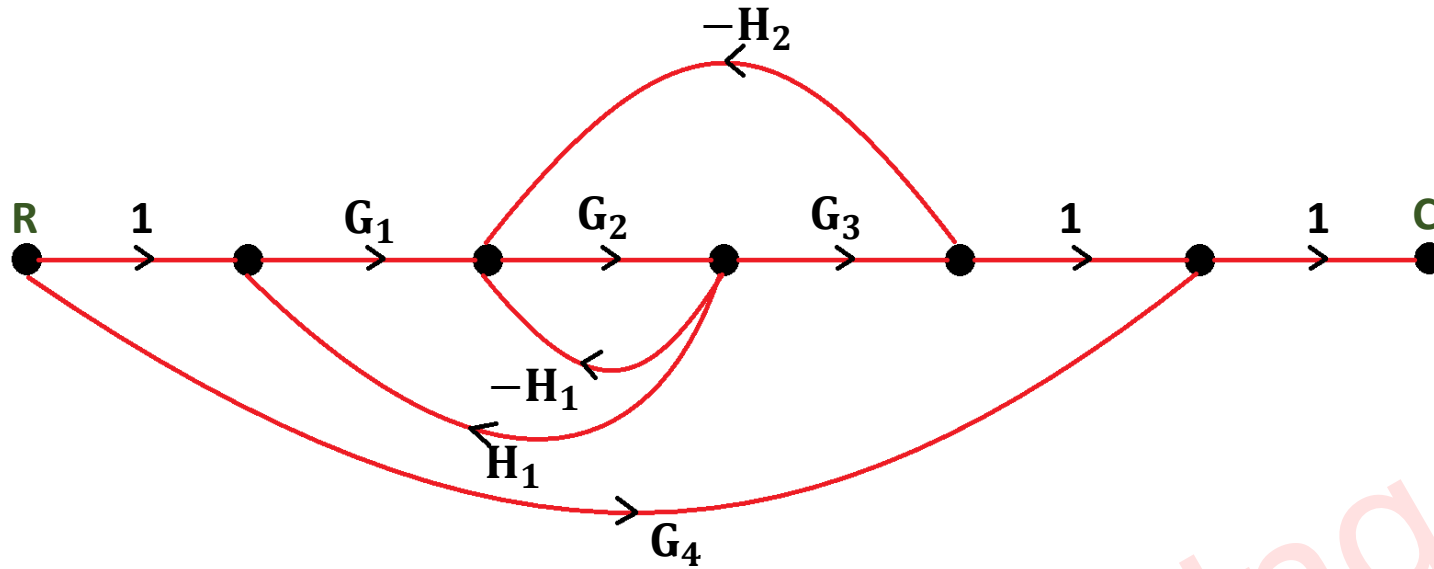
$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-G_1 G_2 G_3 - G_2 - G_3 + G_6 - G_4 G_5) + (-G_1 G_2 G_3 G_6 - G_2 G_6 + G_2 G_4 G_5 - G_3 G_6)$$

$$\Rightarrow \Delta = 1 + (G_1 G_2 G_3 + G_2 + G_3 - G_6 + G_4 G_5) + (-G_1 G_2 G_3 G_6 - G_2 G_6 + G_2 G_4 G_5 - G_3 G_6)$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta}$$

2. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (G_1 G_2 H_1 - G_2 H_1 - G_2 G_3 H_2)$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1 + G_2 G_3 H_2 + G_2 H_1 - G_1 G_2 H_1$$

Find Forward Path

$$\Rightarrow F_1 = 1 \times G_1 \times G_2 \times G_3 \times 1 \times 1 = G_1 G_2 G_3$$

$$\Rightarrow F_2 = G_4 \times 1 = G_4$$

Find Single Loops

$$\Rightarrow L_1 = G_1 \times G_2 \times H_1 = G_1 G_2 H_1$$

$$\Rightarrow L_2 = G_2 \times -H_1 = -G_2 H_1$$

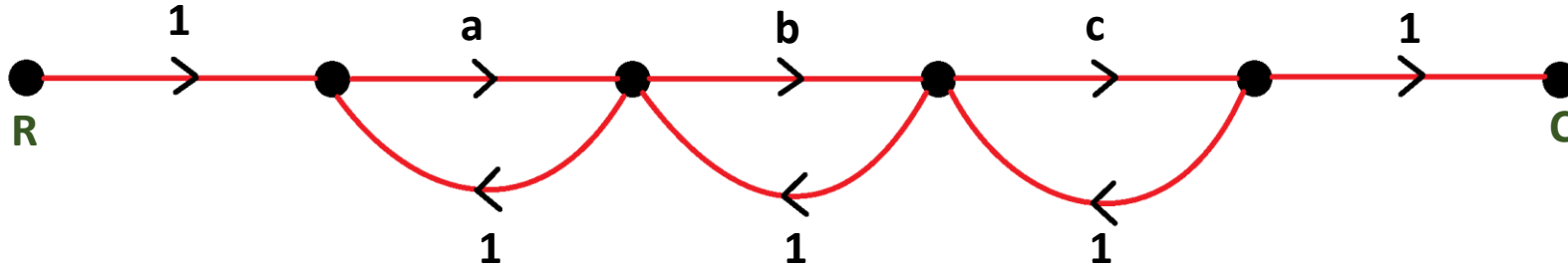
$$\Rightarrow L_3 = G_2 \times G_3 \times -H_2 = -G_2 G_3 H_2$$

There is NO Two Non Touching Loops

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta}$$

$$\Rightarrow \frac{C}{R} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 G_3 H_2 + G_2 H_1 - G_1 G_2 H_1)}{1 + G_2 G_3 H_2 + G_2 H_1 - G_1 G_2 H_1}$$

3. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = 1 \times a \times b \times c \times 1 = abc$$

Find Single Loops

$$\Rightarrow L_1 = a \times 1 = a$$

$$\Rightarrow L_2 = b \times 1 = b$$

$$\Rightarrow L_3 = c \times 1 = c$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_3 = a \times c = ac$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{abc}{1 - (a + b + c) + ac}$$

Find Δ

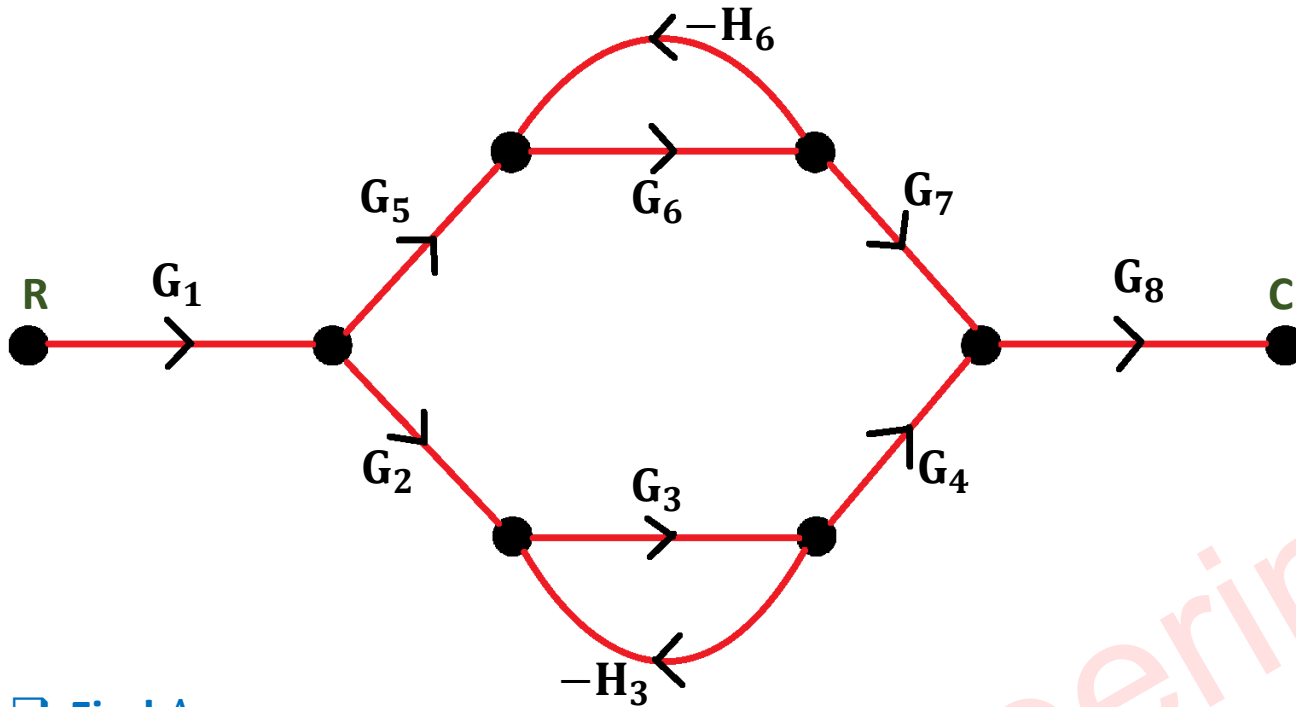
$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (a + b + c) + ac$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

4. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = G_1 \times G_5 \times G_6 \times G_7 \times G_8 = G_1 G_5 G_6 G_7 G_8$$

$$\Rightarrow F_2 = G_1 \times G_2 \times G_3 \times G_4 \times G_8 = G_1 G_2 G_3 G_4 G_8$$

Find Single Loops

$$\Rightarrow L_1 = G_6 \times -H_6 = -G_6 H_6$$

$$\Rightarrow L_2 = G_3 \times -H_3 = -G_3 H_3$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_2 = -G_6 H_6 \times -G_3 H_3 = G_6 G_3 H_3 H_6$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 + (G_6 H_6 + G_3 H_3) + G_6 G_3 H_3 H_6$$

Find Δ_i

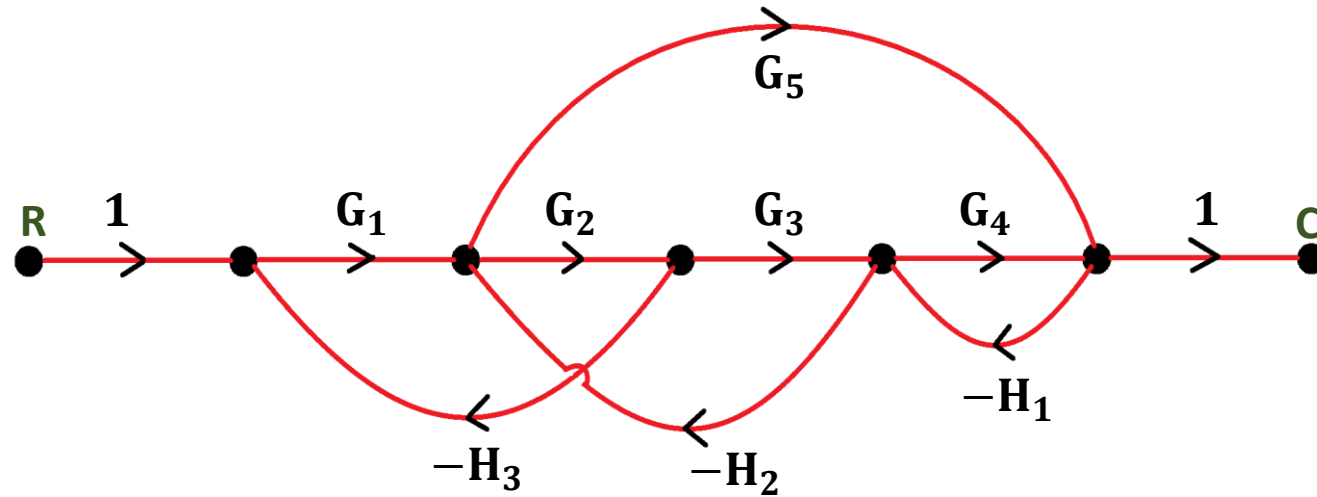
$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1 + G_3 H_3$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1 + G_6 H_6$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta}$$

$$\Rightarrow \frac{C}{R} = \frac{G_1 G_5 G_6 G_7 G_8 (1 + G_3 H_3) + G_1 G_2 G_3 G_4 G_8 (1 + G_6 H_6)}{1 + G_3 H_3 + G_6 H_6 + G_3 G_6 H_3 H_6}$$

5. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-G_1 G_2 H_3 - G_2 G_3 H_2 - G_3 H_1 + G_5 H_1 H_2) + G_1 G_2 G_3 H_1 H_3$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta}$$

$$= \frac{G_1 G_2 G_3 G_4 + G_1 G_5}{1 + G_1 G_2 H_3 + G_2 G_3 H_2 + G_3 H_1 - G_5 H_1 H_2 + G_1 G_2 G_3 H_1 H_3}$$

Find Forward Path

$$\Rightarrow F_1 = 1 \times G_1 \times G_2 \times G_3 \times G_4 \times 1 = G_1 G_2 G_3 G_4$$

$$\Rightarrow F_2 = 1 \times G_1 \times G_5 \times 1 = G_1 G_5$$

Find Single Loops

$$\Rightarrow L_1 = G_1 \times G_2 \times -H_3 = -G_1 G_2 H_3$$

$$\Rightarrow L_2 = G_2 \times G_3 \times -H_2 = -G_2 G_3 H_2$$

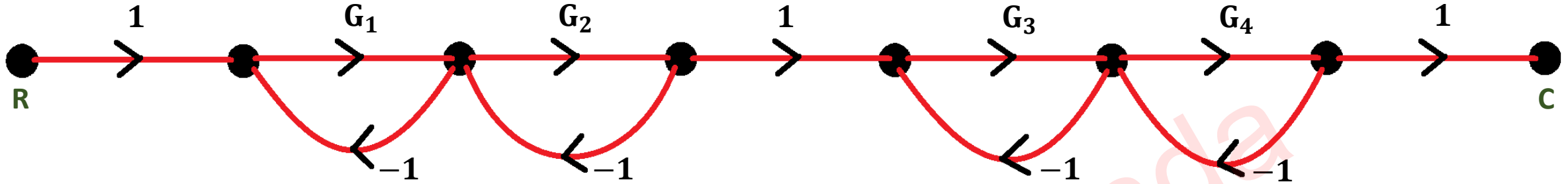
$$\Rightarrow L_3 = G_4 \times -H_1 = -G_4 H_1$$

$$\Rightarrow L_4 = G_5 \times -H_1 \times -H_2 = G_5 H_1 H_2$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_3 = G_1 G_2 G_3 H_1 H_3$$

6. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = G_1 \times G_2 \times G_3 \times G_4 = G_1 G_2 G_3 G_4$$

Find Single Loops

$$\Rightarrow L_1 = -G_1$$

$$\Rightarrow L_2 = -G_2$$

$$\Rightarrow L_3 = -G_3$$

$$\Rightarrow L_4 = -G_4$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_3 = G_1 G_3$$

$$\Rightarrow L_2' = L_1 \times L_4 = G_1 G_4$$

$$\Rightarrow L_3' = L_2 \times L_3 = G_2 G_3$$

$$\Rightarrow L_4' = L_2 \times L_4 = G_2 G_4$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

Find Δ

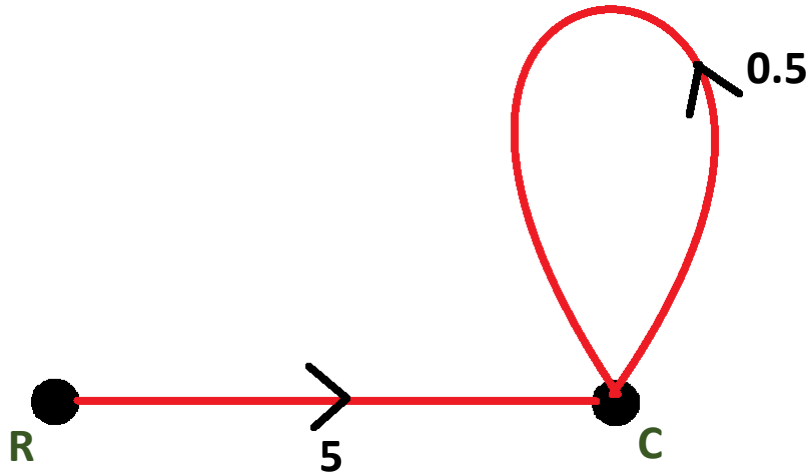
$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 + (G_1 + G_2 + G_3 + G_4) + (G_1 G_3 + G_1 G_4 + G_2 G_3 + G_2 G_4)$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{G_1 G_2 G_3 G_4}{1 + (G_1 + G_2 + G_3 + G_4) + (G_1 G_3 + G_1 G_4 + G_2 G_3 + G_2 G_4)}$$

7. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



□ Find Forward Path

$$\Rightarrow F_1 = 5$$

□ Find Single Loops

$$\Rightarrow L_1 = 0.5$$

□ Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{5 \times 1}{0.5} = 10$$

□ Find Δ_i

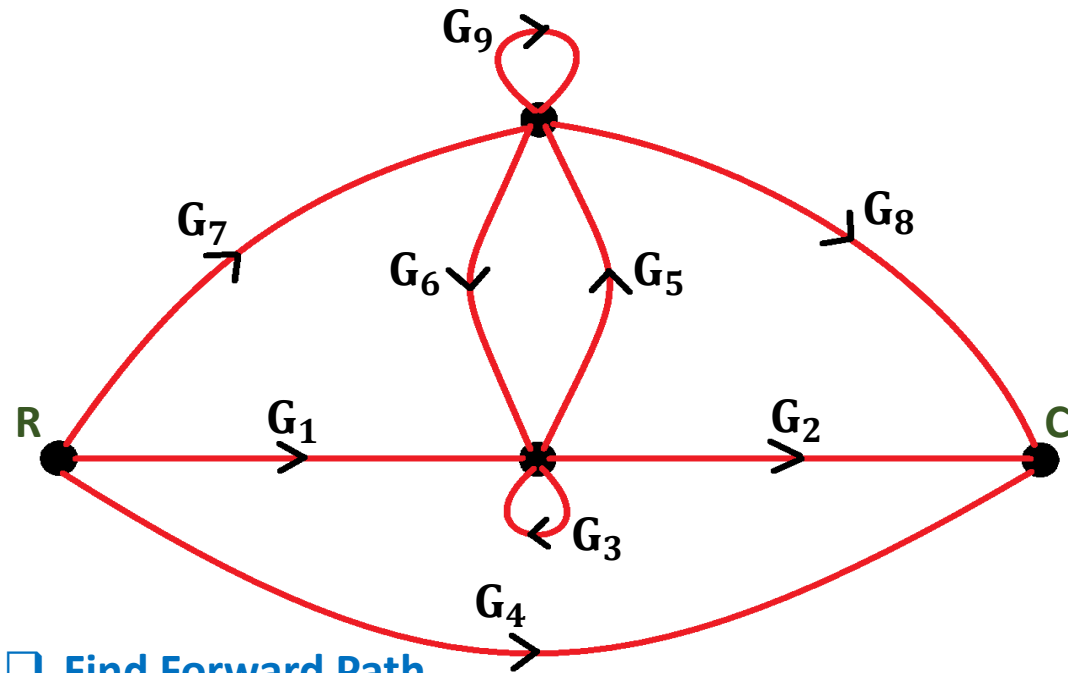
$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

□ Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - 0.5 = 0.5$$

8. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_2 = G_9 G_3$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1 - G_9$$

$$\begin{aligned} \Rightarrow \Delta_2 &= 1 - \sum \text{Single Loops Not touching path } F_2 \\ &= 1 - (G_3 + G_9 + G_5 G_6) \end{aligned}$$

$$\Rightarrow \Delta_3 = 1 - \sum \text{Single Loops Not touching path } F_3 = 1 - G_3$$

$$\Rightarrow \Delta_4 = 1 - \sum \text{Single Loops Not touching path } F_4 = 1$$

$$\Rightarrow \Delta_5 = 1 - \sum \text{Single Loops Not touching path } F_5 = 1$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (G_9 + G_3 + G_6 G_5) + G_9 G_3$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2 + F_3 \Delta_3 + F_4 \Delta_4 + F_5 \Delta_5}{\Delta}$$

Find Forward Path

$$\Rightarrow F_1 = G_1 \times G_2 = G_1 G_2$$

$$\Rightarrow F_2 = G_4$$

$$\Rightarrow F_3 = G_7 \times G_8 = G_7 G_8$$

$$\Rightarrow F_4 = G_1 \times G_5 \times G_8 = G_1 G_5 G_8$$

$$\Rightarrow F_5 = G_7 \times G_6 \times G_2 = G_7 G_6 G_2$$

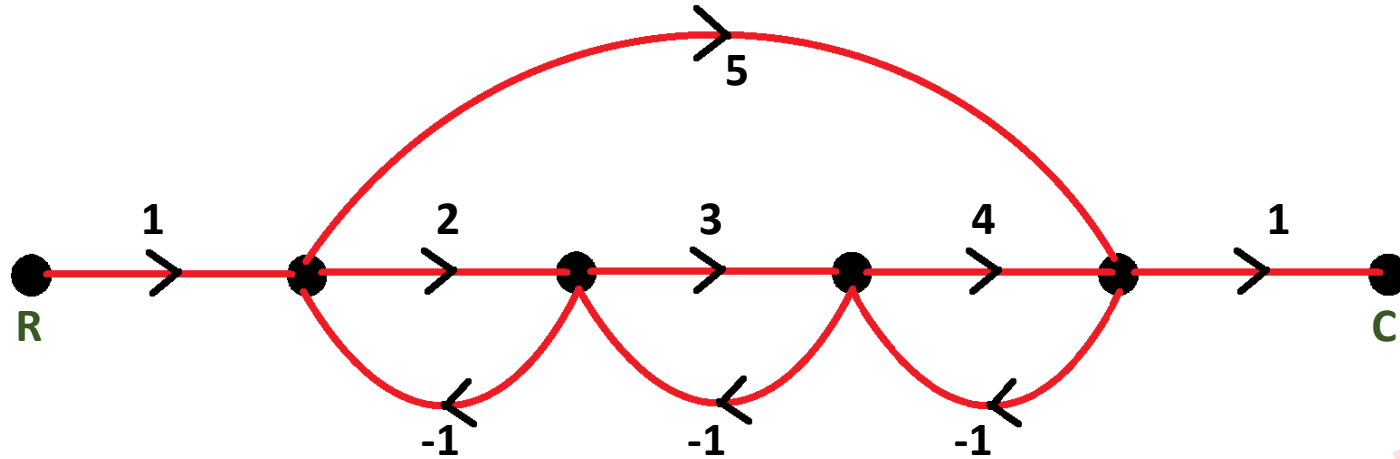
Find Single Loops

$$\Rightarrow L_1 = G_9$$

$$\Rightarrow L_2 = G_3$$

$$\Rightarrow L_3 = G_6 \times G_5 = G_6 G_5$$

9. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = 1 \times 2 \times 3 \times 4 \times 1 = 24$$

$$\Rightarrow F_2 = 1 \times 5 \times 1 = 5$$

Find Single Loops

$$\Rightarrow L_1 = 2 \times -1 = -2$$

$$\Rightarrow L_2 = 3 \times -1 = -3$$

$$\Rightarrow L_3 = 4 \times -1 = -4$$

$$\Rightarrow L_4 = 5 \times -1 \times -1 \times -1 = -5$$

Find Two Non Touching Loops

$$\Rightarrow L_1' = L_1 \times L_3 = 8$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1 - (-3) = 4$$

Find Δ

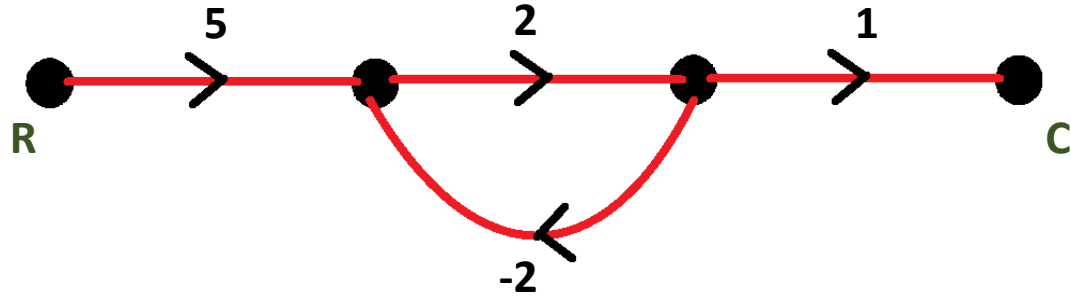
$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-2 - 3 - 4 - 5) + 8 = 23$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta} = \frac{24 \times 1 + 5 \times 4}{23} = \frac{44}{23}$$

10. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = 5 \times 2 \times 1 = 10$$

Find Single Loops

$$\Rightarrow L_1 = 2 \times -2 = -4$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

Find Δ

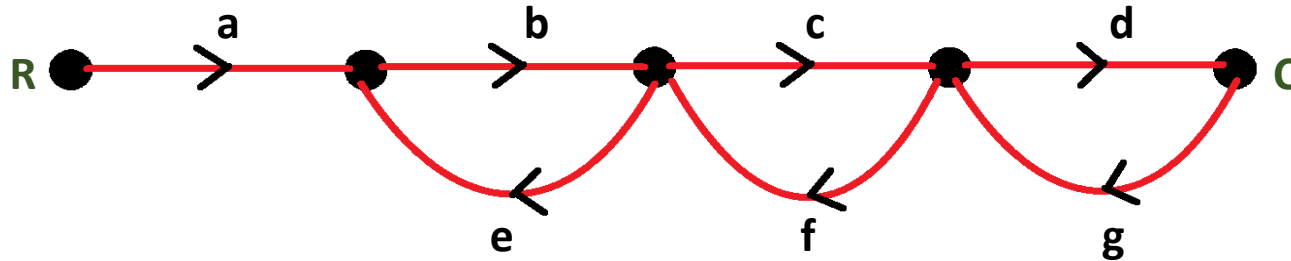
$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-4) = 5$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{10 \times 1}{5} = 2$$

11. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = a \times b \times c \times d = abcd$$

Find Single Loops

$$\Rightarrow L_1 = b \times e = be$$

$$\Rightarrow L_2 = c \times f = cf$$

$$\Rightarrow L_3 = d \times g = dg$$

Find Two Non-Touching Single Loops

$$\Rightarrow L'_1 = L_1 \times L_3 = bedg$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{abcd}{1 - (be + cf + dg) + bedg}$$

Find Δ_i

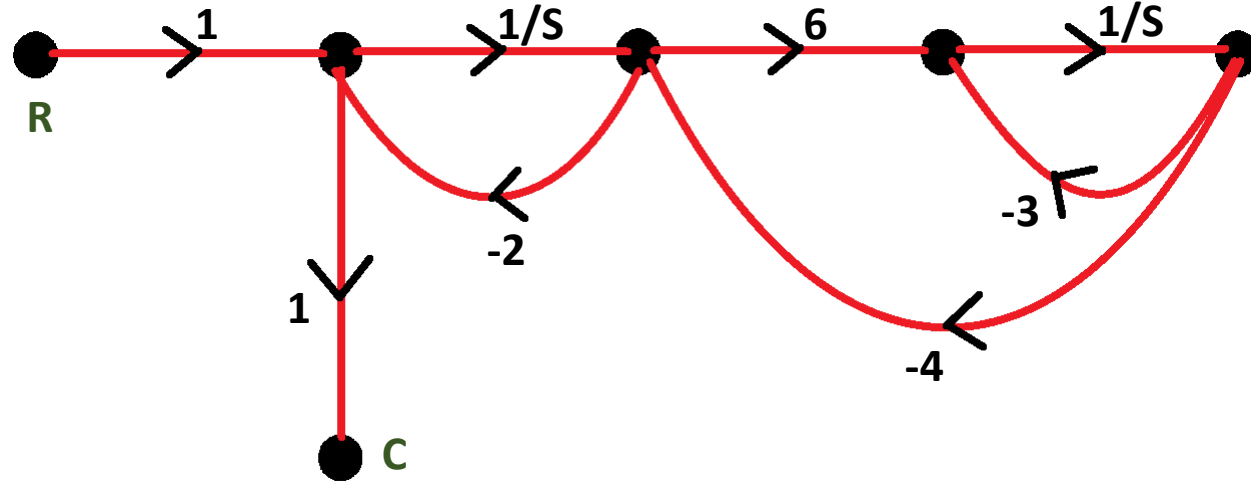
$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (be + cf + dg) + bedg$$

12. Find the Transfer Function of the Signal Flow Graph using the Meson Gain Formula.



Find Forward Path

$$\Rightarrow F_1 = 1 \times 1 = 1$$

Find Single Loops

$$\Rightarrow L_1 = 1/S \times -2 = -2/S$$

$$\Rightarrow L_2 = 1/S \times -3 = -3/S$$

$$\Rightarrow L_3 = 6 \times 1/S \times -4 = -24/S$$

Find Two Non-Touching Single Loops

$$\Rightarrow L'_1 = L_1 \times L_2 = 6/S^2$$

Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1}{\Delta} = \frac{1 + \frac{27}{S}}{1 + \frac{29}{S} + \frac{6}{S^2}} = \frac{S(S + 27)}{S^2 + 29S + 6}$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1 + \frac{3}{S} + \frac{24}{S} = 1 + \frac{27}{S}$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - \left(-\frac{2}{S} - \frac{3}{S} - \frac{24}{S} \right) + \frac{6}{S^2}$$

$$\Rightarrow \Delta = 1 + \frac{29}{S} + \frac{6}{S^2}$$

Signal Flow Graph From Line Equations

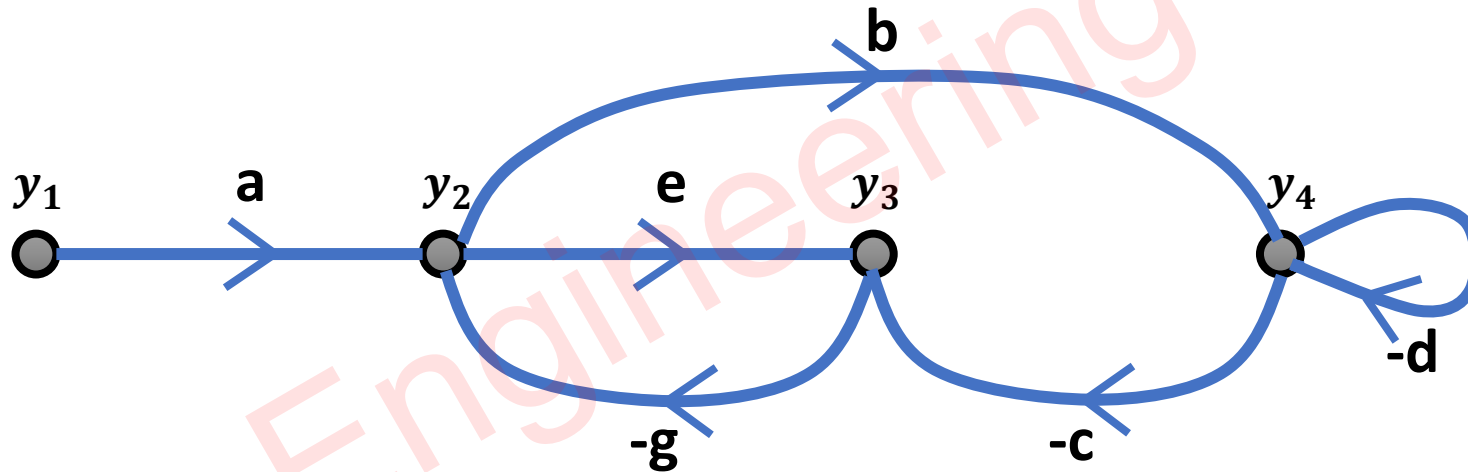
Construct Signal Flow graph from given line equations

$$\Rightarrow y_2 = ay_1 - gy_3$$

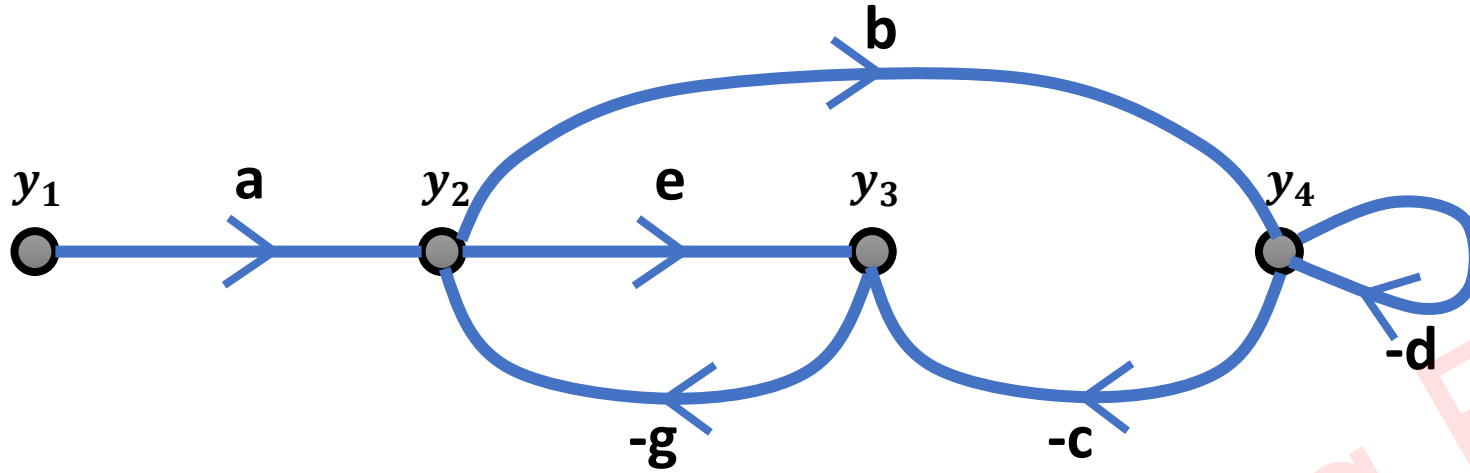
$$\Rightarrow y_3 = ey_2 - cy_4$$

$$\Rightarrow y_4 = by_2 - dy_4$$

Also find transfer function y_4/y_1



Also find transfer function y_4/y_1



□ Find Forward Path

$$\Rightarrow F_1 = a \times b = ab$$

□ Find Single Loops

$$\Rightarrow L_1 = e \times -g = -eg$$

$$\Rightarrow L_2 = b \times -c \times -g = bcg$$

$$\Rightarrow L_3 = -d$$

□ Find Two Non-Touching Single Loops

$$\Rightarrow L'_1 = L_1 \times L_3 = egd$$

□ Transfer Function

$$\Rightarrow \frac{y_4}{y_1} = \frac{F_1 \Delta_1}{\Delta} = \frac{ab}{1 + eg - bcg + d + egd}$$

□ Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

□ Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-eg + bcg - d) + egd$$

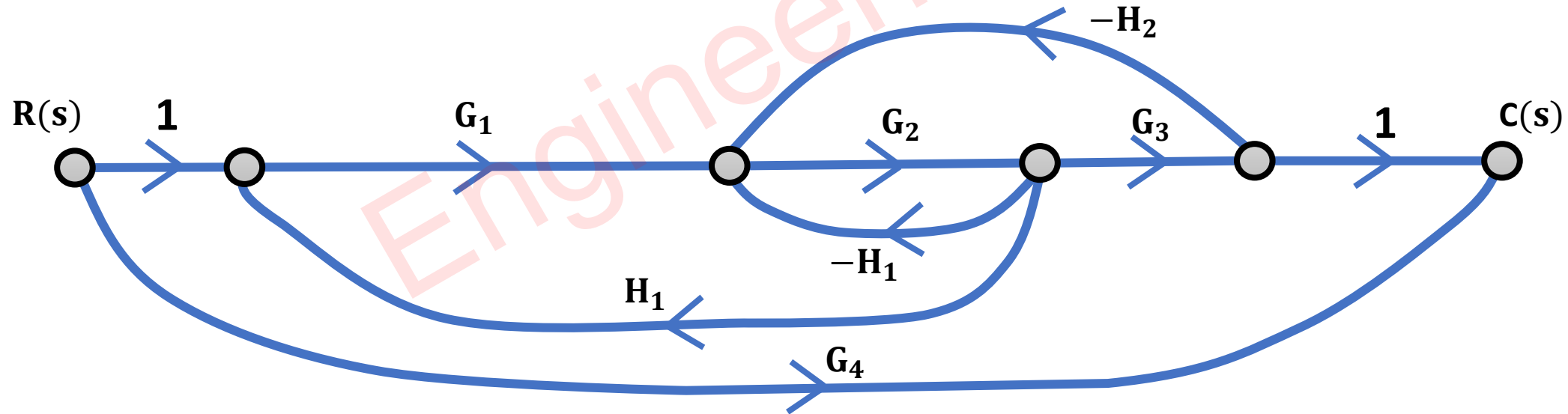
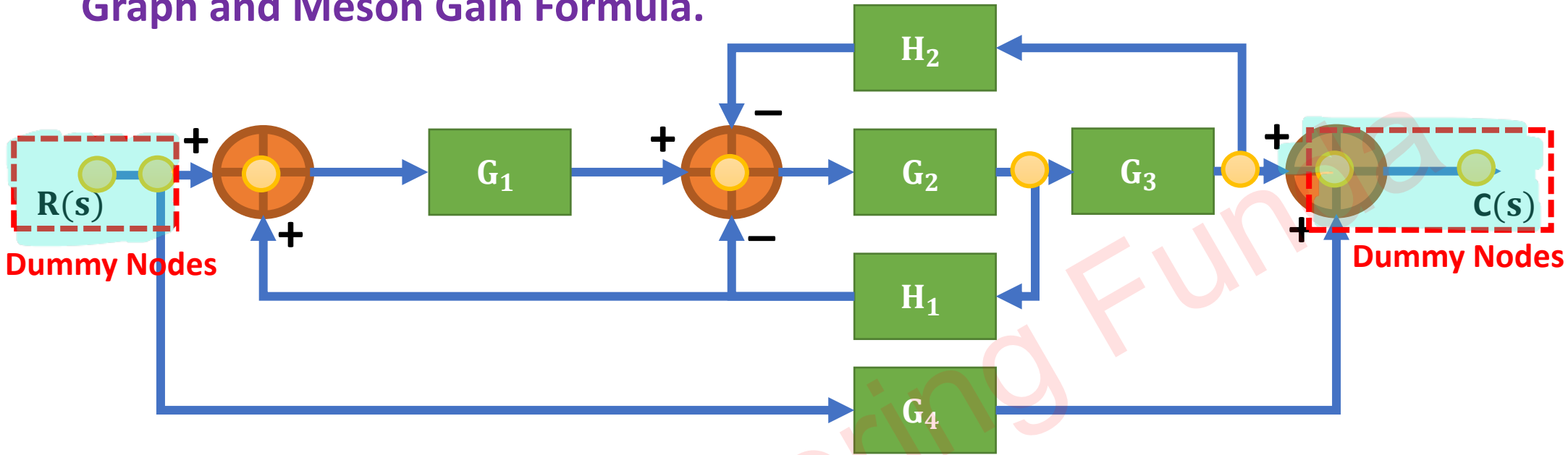
$$\Rightarrow \Delta = 1 + eg - bcg + d + egd$$

Block Diagram to Signal Flow Graph

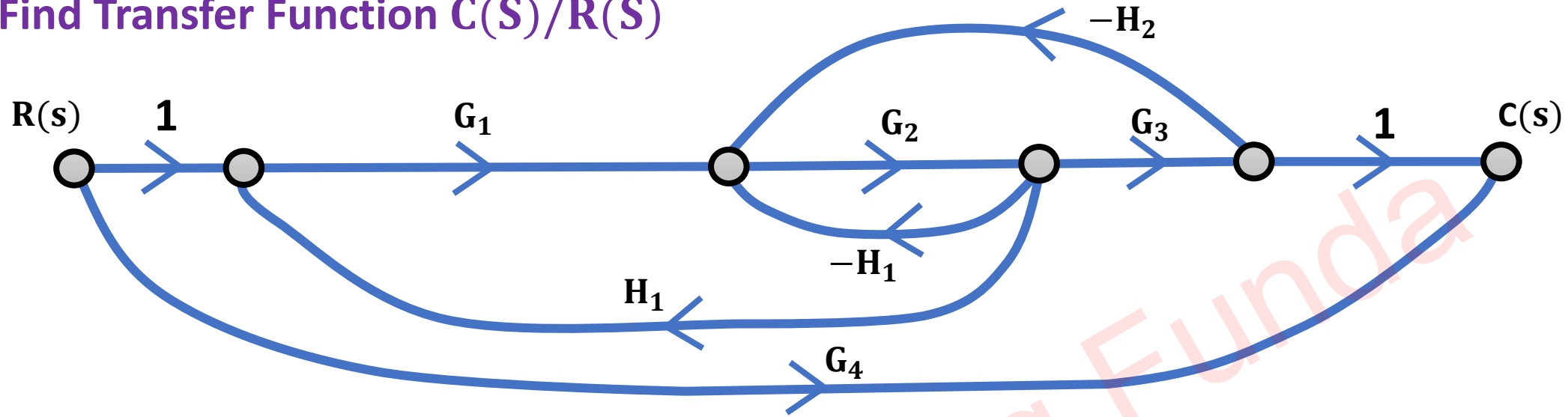
Steps of Block Diagram to Signal Flow Graph Conversion

- ☐ Node Assignment to Input, Output, All Adders, All Joints, and between Two Consecutive Blocks.
- ☐ Eliminate Dummy Nodes.
- ☐ Join All nodes with Direction and Gain.

1. Find the Transfer Function of the given Block Diagram using Conversion to Signal Flow Graph and Meson Gain Formula.



Find Transfer Function $C(S)/R(S)$



Find Forward Path

$$\Rightarrow F_1 = G_1 \times G_2 \times G_3 = G_1 G_2 G_3$$

$$\Rightarrow F_2 = G_4$$

Find Single Loops

$$\Rightarrow L_1 = G_2 \times -H_1 = -G_2 H_1$$

$$\Rightarrow L_2 = G_1 \times G_2 \times H_1 = G_1 G_2 H_1$$

$$\Rightarrow L_3 = G_2 \times G_3 \times -H_2 = -G_2 G_3 H_2$$

No Two Non-Touching Loops

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1 + G_2 H_1 - G_1 G_2 H_1 + G_2 G_3 H_2$$

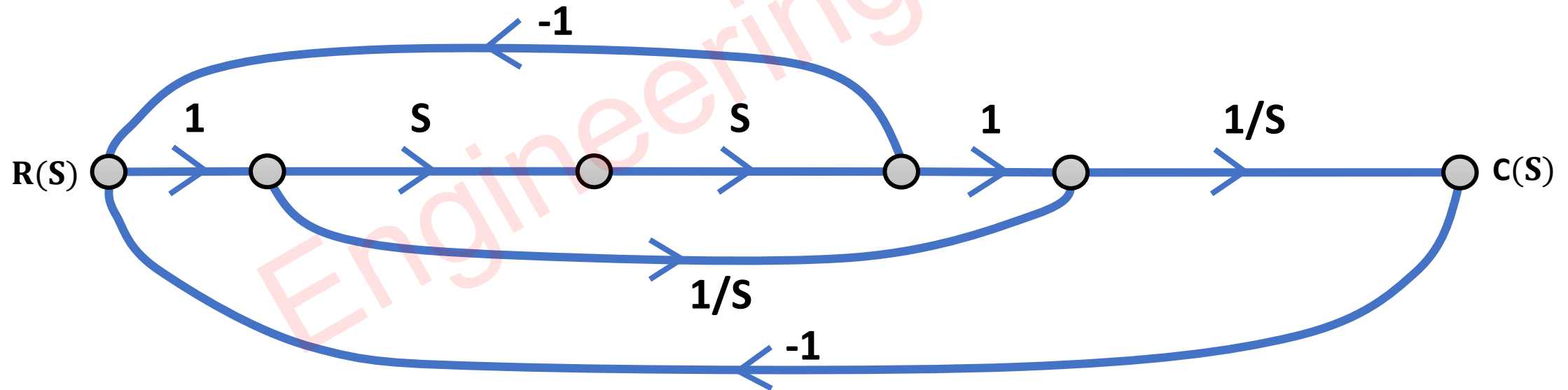
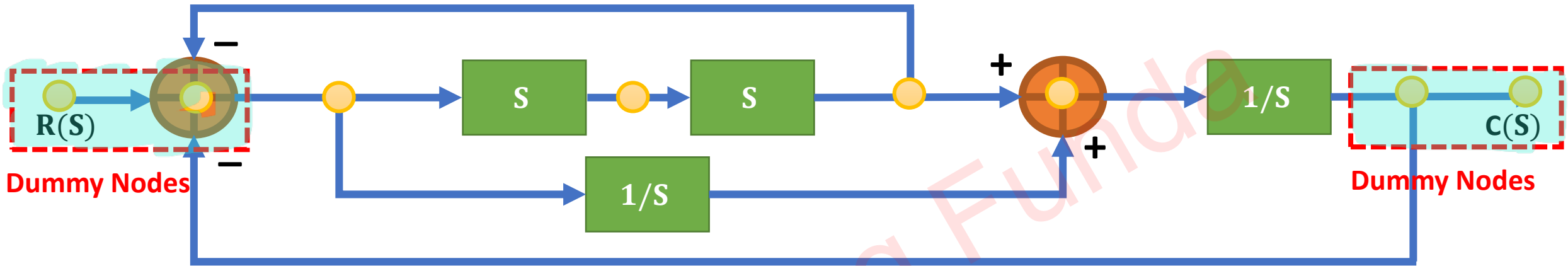
Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

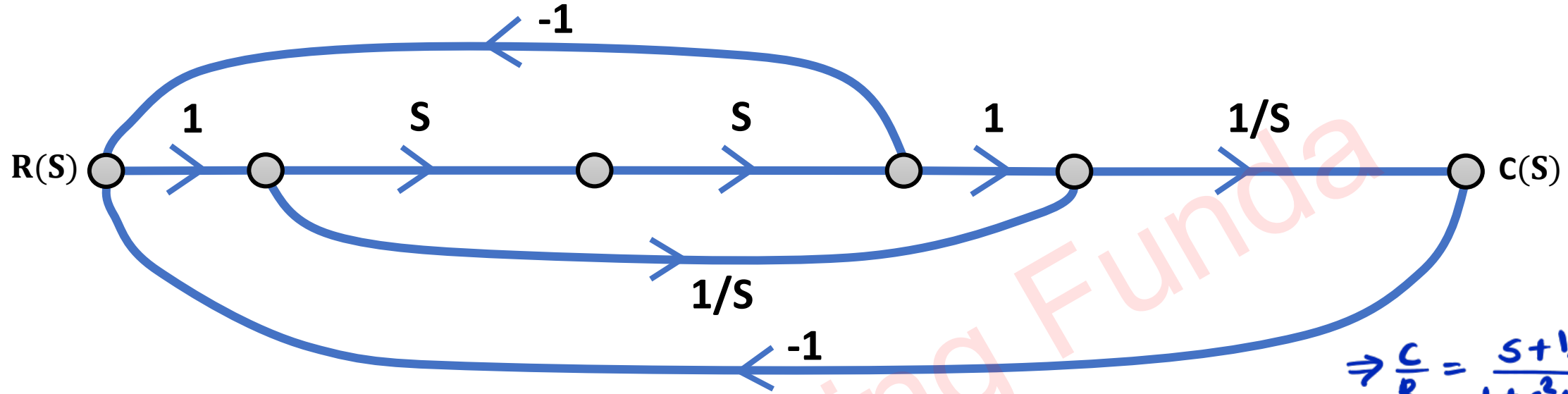
$$\Rightarrow \Delta = 1 + G_2 H_1 - G_1 G_2 H_1 + G_2 G_3 H_2$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 H_1 - G_1 G_2 H_1 + G_2 G_3 H_2)}{1 + G_2 H_1 - G_1 G_2 H_1 + G_2 G_3 H_2}$$

2. Find the Transfer Function of the given Block Diagram using Conversion to Signal Flow Graph and Meson Gain Formula.



Find Transfer Function $C(S)/R(S)$



$$\Rightarrow \frac{C}{R} = \frac{S + 1/S^2}{1 + S^2 + S + 1/S^2}$$

$$= \frac{S^3 + 1}{S^4 + S^3 + S^2 + 1}$$

Find Forward Path

$$\Rightarrow F_1 = 1 \times S \times S \times 1 \times 1/S = S$$

$$\Rightarrow F_2 = 1 \times 1/S \times 1/S = 1/S^2$$

Find Single Loops

$$\Rightarrow L_1 = 1 \times S \times S \times -1 = -S^2$$

$$\Rightarrow L_2 = 1 \times S \times S \times 1 \times \frac{1}{S} \times -1 = -S$$

$$\Rightarrow L_3 = 1 \times \frac{1}{S} \times \frac{1}{S} \times -1 = \frac{-1}{S^2}$$

No Two Non-Touching Loops

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1$$

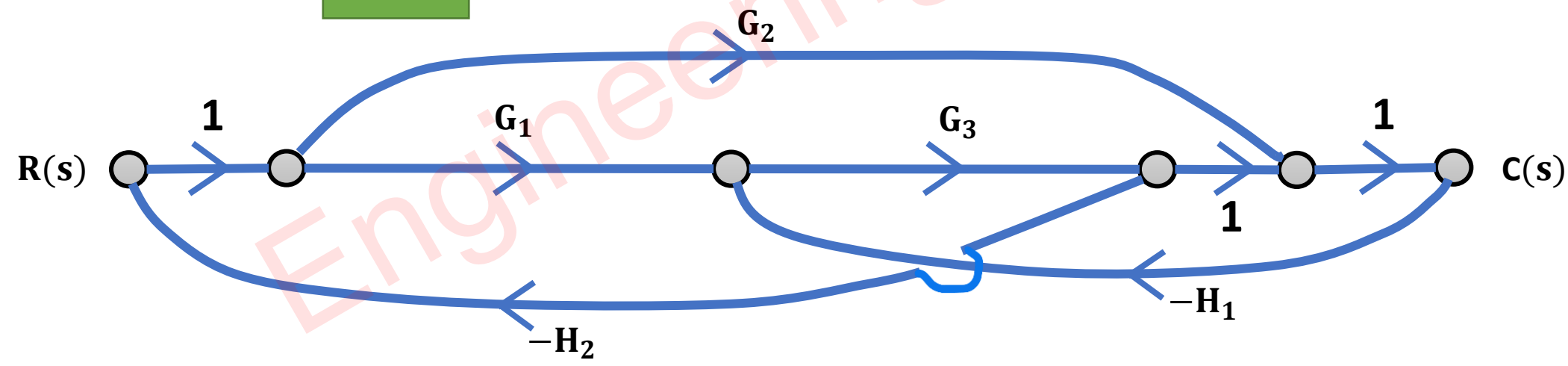
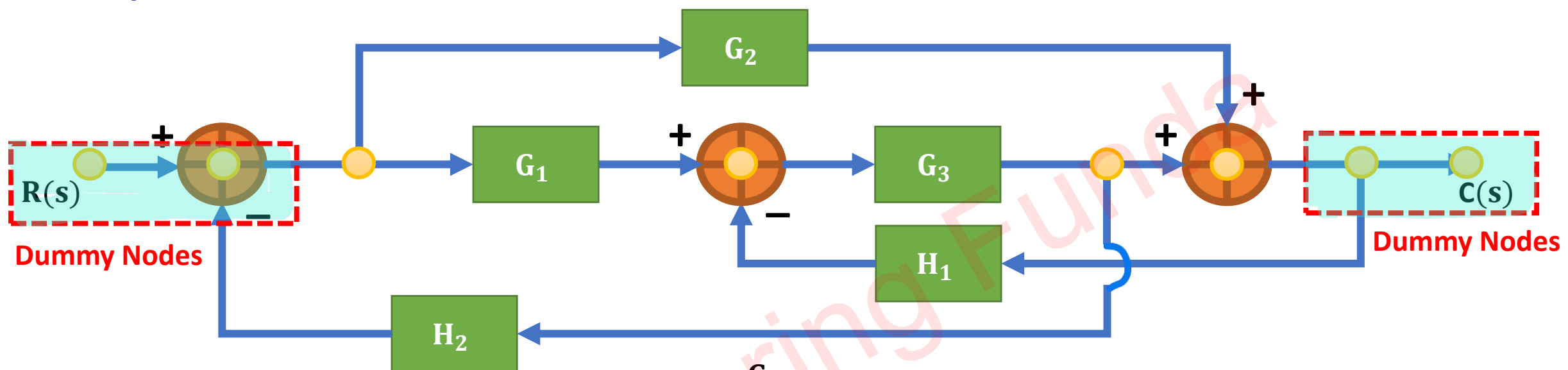
Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

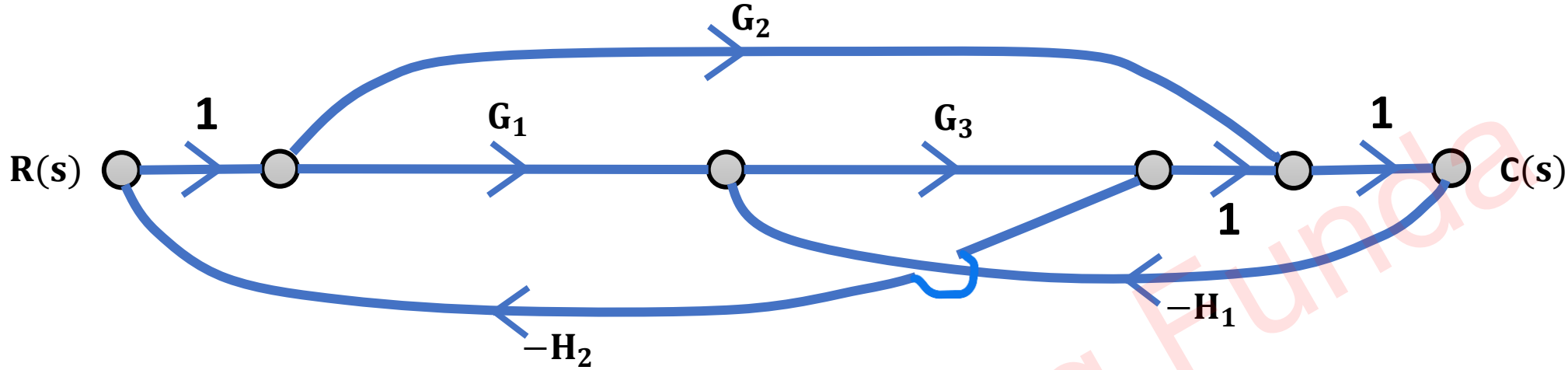
$$\Rightarrow \Delta = 1 - \left(-S^2 - S - \frac{1}{S^2} \right) = 1 + S^2 + S + \frac{1}{S^2}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta}$$

3. Find the Transfer Function of the given Block Diagram using Conversion to Signal Flow Graph and Meson Gain Formula.



Find Transfer Function $C(S)/R(S)$



Find Forward Path

$$\Rightarrow F_1 = 1 \times G_1 \times G_3 \times 1 \times 1 = G_1 G_3$$

$$\Rightarrow F_2 = 1 \times G_2 \times 1 = G_2$$

Find Single Loops

$$\Rightarrow L_1 = 1 \times G_1 \times G_3 \times -H_2 = -G_1 G_3 H_2$$

$$\Rightarrow L_2 = G_3 \times 1 \times 1 \times -H_1 = -G_3 H_1$$

$$\Rightarrow L_3 = 1 \times G_2 \times 1 \times -H_1 \times G_3 \times -H_2 = G_2 G_3 H_1 H_2$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

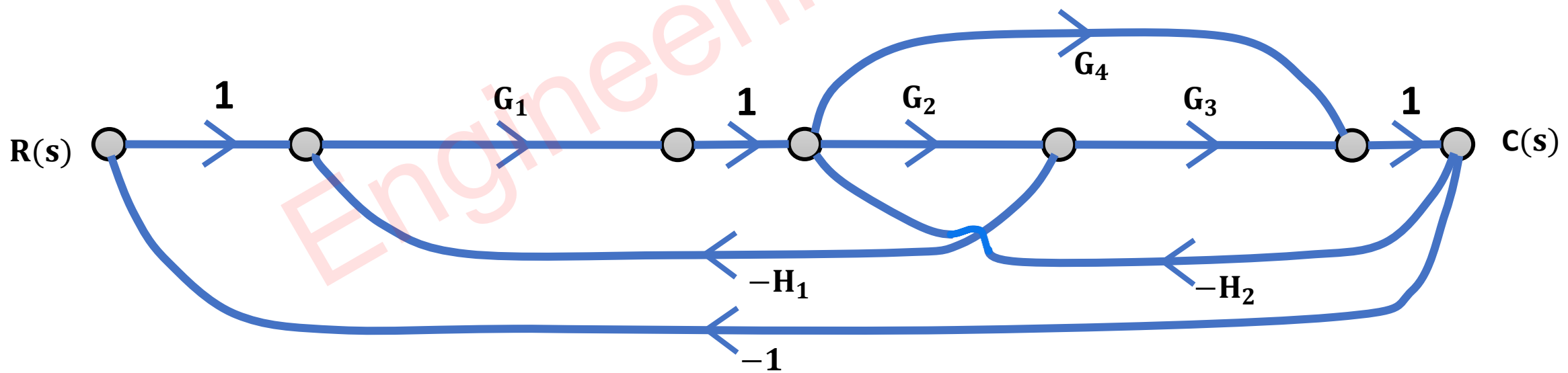
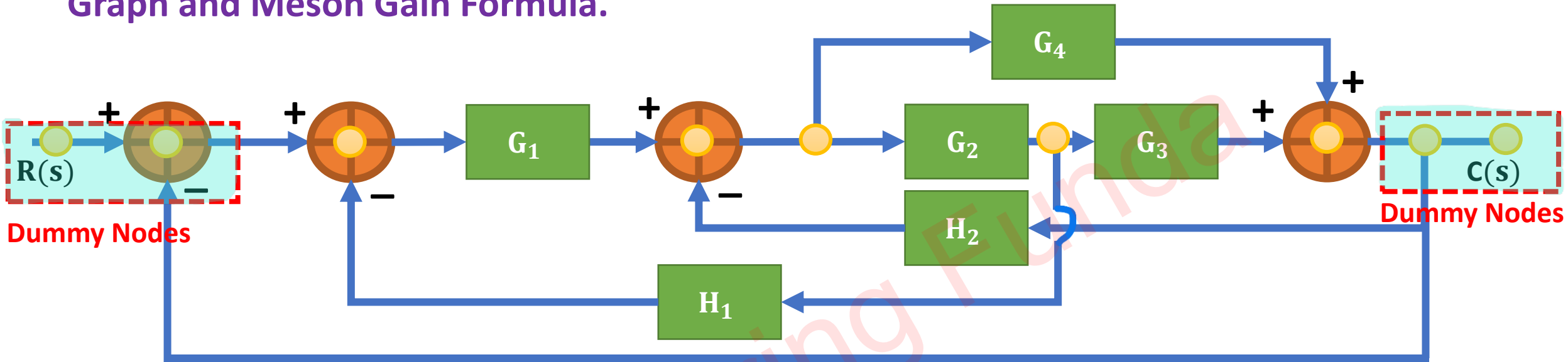
$$\Rightarrow \Delta = 1 - (-G_1 G_3 H_2 - G_3 H_1 + G_2 G_3 H_1 H_2)$$

$$\Rightarrow \Delta = 1 + G_1 G_3 H_2 + G_3 H_1 - G_2 G_3 H_1 H_2$$

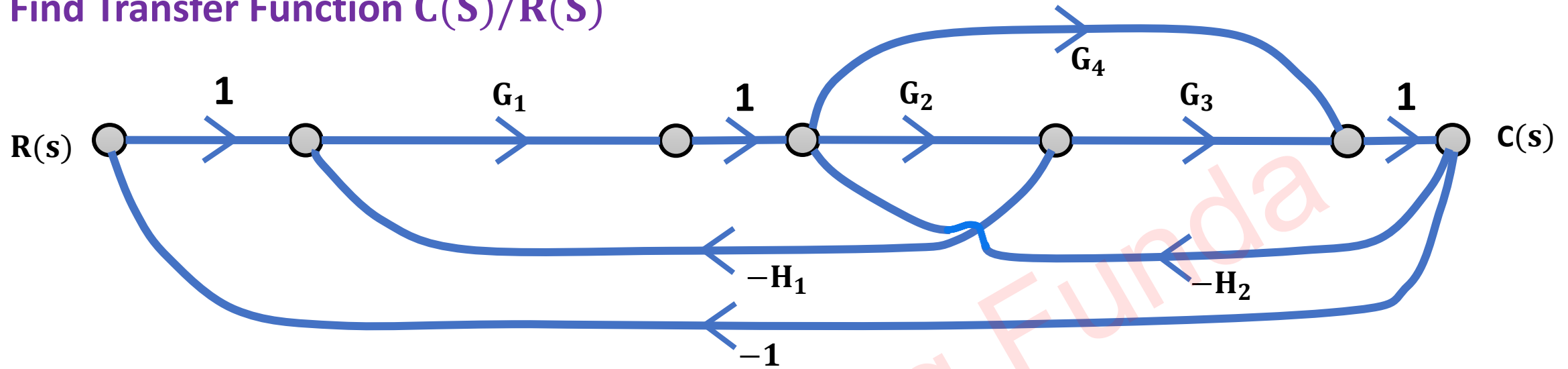
Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta} = \frac{G_1 G_3 + G_2}{1 + G_1 G_3 H_2 + G_3 H_1 - G_2 G_3 H_1 H_2}$$

4. Find the Transfer Function of the given Block Diagram using Conversion to Signal Flow Graph and Meson Gain Formula.



Find Transfer Function $C(S)/R(S)$



Find Forward Path

$$\Rightarrow F_1 = 1 \times G_1 \times 1 \times G_2 \times G_3 \times 1 = G_1 G_2 G_3$$

$$\Rightarrow F_2 = 1 \times G_1 \times 1 \times G_4 \times 1 = G_1 G_4$$

Find Single Loops

$$\Rightarrow L_1 = G_1 \times 1 \times G_2 \times -H_1 = -G_1 G_2 H_1$$

$$\Rightarrow L_2 = G_2 \times G_3 \times 1 \times -H_2 = -G_2 G_3 H_2$$

$$\Rightarrow L_3 = G_4 \times 1 \times -H_2 = -G_4 H_2$$

$$\Rightarrow L_4 = 1 \times G_1 \times 1 \times G_2 \times G_3 \times 1 \times -1 = -G_1 G_2 G_3$$

$$\Rightarrow L_5 = 1 \times G_1 \times 1 \times G_4 \times 1 \times -1 = -G_1 G_4$$

Find Δ_i

$$\Rightarrow \Delta_1 = 1 - \sum \text{Single Loops Not touching path } F_1 = 1$$

$$\Rightarrow \Delta_2 = 1 - \sum \text{Single Loops Not touching path } F_2 = 1$$

Find Δ

$$\Rightarrow \Delta = 1 - \sum \text{Single Loops} + \sum \text{Two Non Touching Loops}$$

$$\Rightarrow \Delta = 1 - (-G_1 G_2 H_1 - G_2 G_3 H_2 - G_4 H_2 - G_1 G_2 G_3 - G_1 G_4)$$

$$\Rightarrow \Delta = 1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_4 H_2 + G_1 G_2 G_3 + G_1 G_4$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{F_1 \Delta_1 + F_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_4 H_2 + G_1 G_2 G_3 + G_1 G_4}$$